

NAME (PRINT):

Last/Surname

First /Given Name

STUDENT #:

UTORID:

SIGNATURE:

UNIVERSITY OF TORONTO MISSISSAUGA
APRIL 2023 FINAL EXAMINATION
CSC343H5S
Introduction to Databases
Michael Liut
Duration - 2 hours
Aids: None

The University of Toronto Mississauga and you, as a student, share a commitment to academic integrity. You are reminded that you may be charged with an academic offence for possessing any unauthorized aids during the writing of an exam. Clear, sealable, plastic bags have been provided for all electronic devices with storage, including but not limited to: cell phones, smart watches, SMART devices, tablets, laptops, and calculators. Please turn off all devices, seal them in the bag provided, and place the bag under your desk for the duration of the examination. You will not be able to touch the bag or its contents until the exam is over.

If, during an exam, any of these items are found on your person or in the area of your desk other than in the clear, sealable, plastic bag, you may be charged with an academic offence. A typical penalty for an academic offence may cause you to fail the course.

*Please note, once this exam has begun, you **CANNOT** re-write it.*

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NAME: _____

UTORID: _____

CSC 343 H5SDuration: 2 hours

University of Toronto Mississauga

Final Examination

Instructor: Michael Liut

April 14, 2023

Read all instructions before starting the exam

This test paper includes 19 pages and a total of 20 questions; each question requires a written answer. You are responsible for ensuring that your copy of the paper is complete. Bring any discrepancy to the attention of your invigilator.

Special Instructions:

1. The answers to the true-or-false questions (Questions 1-15) and written questions (Questions 16-20) must be answered in the space provided (i.e. immediately following the question). Written answers are to be clear and concise; illegible solutions will not be marked.
2. Read all questions before starting the exam - some are easier than others. **Chen's Notation is on Page 18.**
3. **This is a closed book exam.** No memory aids or textbooks of any kind are allowed during the test. The use of any electronic device is strictly prohibited. Failure to comply will result in your immediate dismissal from the examination and a grade of 0.
4. You are not allowed to be involved in any communication of any kind concerning the questions and answers of this exam with anybody except the invigilator(s) or the instructor(s). Any attempt of such communication will be considered a case of academic dishonesty.
5. Intentionally blank pages can be used for rough work or if you run out of space when answering questions, just be sure to clearly indicate under the question which additional pages were used to solve the problem(s).
6. Replace the word blank in the following statement correctly (write the answer to the right of this sentence) for a bonus: A blank occurs when two or more transactions are waiting for each other to release a resource, causing both to be unable to proceed.
7. Documents to be returned: this questionnaire (i.e., the exam) and, if used, all scrap paper. All documents must bear the student's name, student's number, and student's UTORID. **If you detach your exam, you must write your name, student number, and UTORID on every page.** Otherwise, if you do not detach the exam, you need only write your information on the front page of this exam.

question	mark	out of
1-15		15
16		3
17		3
18		6
19		7
20		6
total		40

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Question 1 – 15 are True-or-False questions. You are required to illustrate the validity by example/definition, or disprove the validity by a counter-example/definition.

For each question only select one answer (i.e. True or False). To select an answer, you must circle “A. True.” or “B. False.”. A negative marking scheme is not used. Incorrect or missing answers earn a mark of 0. Therefore, do not leave any questions blank. You must justify your answer, do this in the space immediately below the question.

Question 1 [1 mark] A relation $R(A,B,C)$ may have at most one minimal superkey.

- A. True.
- B. False.

Question 2 [1 mark] A weak entity-set is an entity-set that cannot be uniquely identified by its attributes alone; therefore, it must use its primary key in conjunction with a foreign key to uniquely identify tuples.

- A. True.
- B. False.

Question 3 [1 mark] Let's assume a weak entity-set E has a relationship with a supporting entity-set R . The relationship between E and R must be one-to-one.

- A. True.
- B. False.

Question 4 [1 mark] Given two relations, $R(A,B)$ and $S(B,C)$, $\sigma_{A=B}(R \times S) \equiv (R \bowtie_{A=B} S)$.

- A. True.
- B. False.

Question 5 [1 mark] Given the following two relations:

<u>Student</u>				
<u>StudentID</u>	Name	ProgramOfStudy	cGPA	Hobbies
001	John Smith	Computer Science	3.7	Programming, Gaming, Basketball
002	Emily Davis	Psychology	3.9	Reading, Hiking, Painting
003	Michael Lee	Mechanical Engineering	3.2	Soccer, Woodworking, Photography
004	Sarah Jones	Biology	3.8	Writing, Yoga, Traveling
005	David Kim	Economics	3.6	Guitar, Cooking, Running
006	Jennifer Lee	English Literature	3.9	Poetry, Theater, Painting
007	Daniel Nguyen	Chemistry	3.5	Chess, Hiking, Playing Piano
008	Jessica Kim	Business Administration	3.4	Volleyball, Photography, Scrapbooking

<u>Enrolled</u>		
<u>StudentID</u>	<u>CourseID</u>	<u>Instructor Name</u>
001	101	Dr. Johnson
001	201	Dr. Lee
002	101	Dr. Johnson
003	301	Dr. Hernandez
003	401	Dr. Kim
004	201	Dr. Lee
005	101	Dr. Johnson
006	401	Dr. Kim

both relations are in second normal form (2NF).

- A. True.
- B. False.

Question 6 [1 mark] Given the **Student** and **Enrolled** relations in Question 5, this query **Query** generates the following resulting table (i.e., **Result**).

Query:

```

SELECT s.Name, COUNT(*) AS NumCourses
FROM Students s
INNER JOIN Enrolled e ON s.StudentID = e.StudentID
WHERE s.ProgramOfStudy = 'Computer Science'
AND e.CourseID NOT IN (
    SELECT CourseID FROM Enrolled WHERE StudentID = s.StudentID
)
GROUP BY s.Name
HAVING COUNT(*) >= 2;

```

<u>Result</u>	
Name	NumCourses
John Smith	2
Emily Davis	2

- A. True.
- B. False.

Question 7 [1 mark] Given the **Student** relation in Question 5, we can generate the following functional dependencies:

```

StudentID -> Name, ProgramOfStudy, cGPA
Name, ProgramOfStudy -> StudentID
StudentID -> Hobbies

```

- A. True.
- B. False.

Question 8 [1 mark] Given the **Student** and **Enrolled** relations in Question 5, an example of a domain-level constraint would be that **Student.cGPA** must be a real data type.

- A. True.
- B. False.

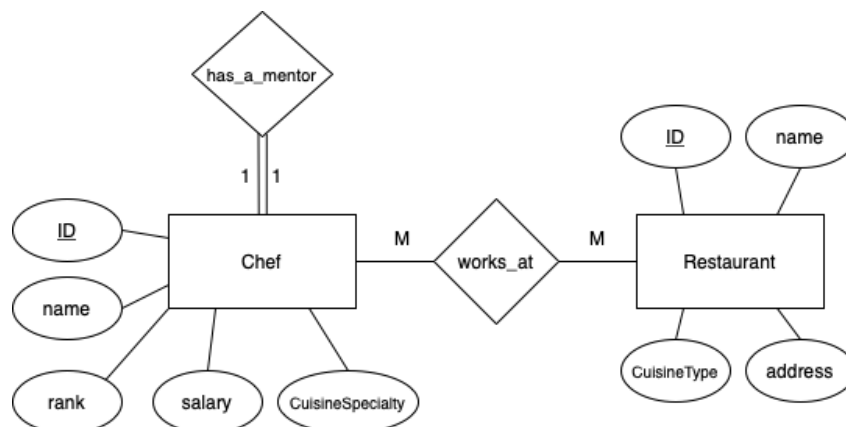
Question 9 [1 mark] Given the **Student** and **Enrolled** relations in Question 5, if you take **Student** and perform a right outer join on **Enrolled** with the following query:

```
SELECT *
FROM Enrolled
RIGHT OUTER JOIN Student
ON Enrolled.StudentID = Student.StudentID;
```

the resulting table will have a total of two NULL values.

- A. True.
- B. False.

Question 10 [1 mark] Given the following ERD:



it is clear that a restaurant can exist without having a chef working there.

- A. True.
- B. False.

Question 11 [1 mark] Given the ERD in Question 10, every chef must have a mentor, thus, every chef must also be a mentor to another chef. The only exception is when there are an odd number of chef records.

- A. True.
- B. False.

Question 12 [1 mark] SQL injection attacks can only be performed by exploiting user input fields on a website.

- A. True.
- B. False.

Question 13 [1 mark] In database normalization, it is always necessary to achieve the highest level of normalization, known as Boyce-Codd Normal Form (BCNF).

- A. True.
- B. False.

Question 14 [1 mark] Given two relations, **Parent** and **Child**, specifying **ON DELETE RESTRICT** will ensure that any deletion of a row in the **Parent** table is automatically deleted from the **Child** table too.

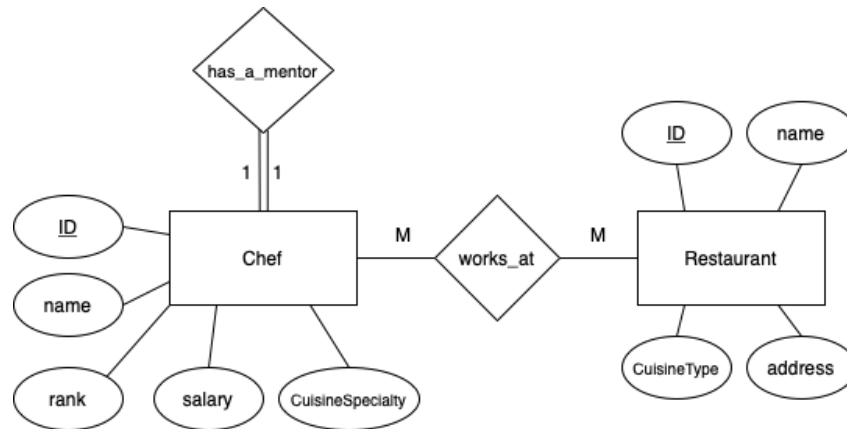
- A. True.
- B. False.

Question 15 [1 mark] Transactions are used to group multiple database operations into a single atomic unit that either completes successfully or fails completely. However, transactions are only used in high-concurrency database environments where multiple users are accessing and updating the same data concurrently.

- A. True.
- B. False.

Questions 16 – 20 are questions that require a written answer. The answers are to be written in the space provided below each question.

Question 16 [3 marks] Given the following ERD:



write the relational algebra which finds the average salary of all chefs whose cuisine speciality is “Italian” but the Restaurant’s cuisine type is not “Italian”. Return the Restaurant’s ID and the average salary of those chefs.

Question 17 [3 marks] From the ERD given in Question 16, write the relational algebra which finds all chefs whose mentors also work at the same restaurant as them. Return the Restaurant’s ID and the count (i.e., the count of the number of chefs whose mentors also work at the same restaurant).

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Question 18 [6 marks] Consider a relation R with a set of attributes $\alpha = \{A, B, C, D, E, F\}$ and the set of Functional Dependencies $\mathcal{F} = \{ A \rightarrow D, D \rightarrow E, AC \rightarrow F, BF \rightarrow E, DE \rightarrow A, EC \rightarrow D \}$, complete the following tasks:

- (a) find all the candidate keys (i.e., minimal keys) of relation R ; and
- (b) decompose this relation R into BCNF. If it is already in BCNF, or once it is, explain why.

Question 19 [7 marks] Consider a relation **Project**(projectnum, projecttitle, projectmgr, phone), where $\text{projectmgr} \rightarrow \text{phone}$.

1. Show that **Project** is not in 3NF.
2. Does the fact that **Project** is not in 3NF lead to insertion anomaly? If so, give an example.
3. Does the fact that **Project** is not in 3NF lead to update anomaly? If so, give an example.
4. Does the fact that **Project** is not in 3NF lead to deletion anomaly? If so, give an example.
5. Decompose **Project** to two relations called **Project1** and **Manager** that are in 3NF. The decomposition must be lossless.

Question 20 [6 marks] For each of the following schedules below (a – e), identify which of these desirable properties are ensured: *serializable*, *conflict-serializable*, *recoverable*, and *avoids-cascading-aborts*.

The actions are listed in the order they are scheduled and prefixed with the transaction name. If a commit or abort is not shown, the schedule is incomplete; assume that abort or commit must follow all the listed actions.

- (a) $T_1:W(A), T_2:R(B), T_1:R(B), T_2:R(A)$
- (b) $T_1:R(A), T_2:W(A), T_1:W(A), T_2:Abort, T_1:Commit$
- (c) $T_1:W(A), T_2:R(A), T_1:W(A), T_2:Abort, T_1:Commit$
- (d) $T_2:R(A), T_3:W(A), T_3:Commit, T_1:W(B), T_1:Commit, T_2:R(B), T_2:W(C), T_2:Commit$
- (e) $T_1:R(A), T_2:W(A), T_2:Commit, T_1:W(A), T_1:Commit, T_3:R(A), T_3:Commit$

Answers are to be written in the table below; you are to use a check mark “✓” to identify which desirable properties are guaranteed and an “X” to identify the ones which are not guaranteed. For those that cannot be determined mark them with a “?” and add an explanation (below the table).

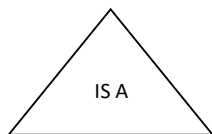
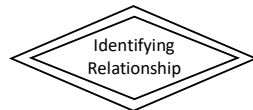
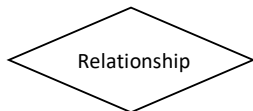
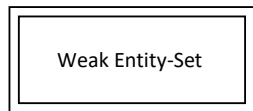
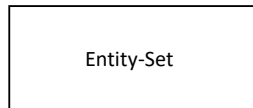
Property Question	Serializable	Conflict-Serializable	Recoverable	Avoids Cascading Aborts
(a)				
(b)				
(c)				
(d)				
(e)				

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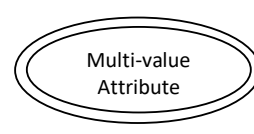
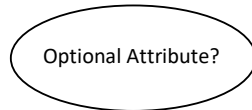
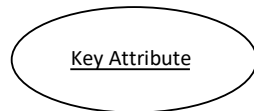
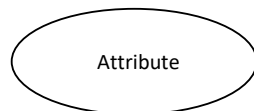
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ERD Cheat Sheet “Chen’s Notation”



Hierarchical Construct



Participations
Cardinality can be shown or hidden

Mandatory

_____ 1 (0:1)

1 _____ 1 (1:1)

_____ M (0:M)

1 _____ M (1:M)

Recursive Relationship
Cardinality can be shown or hidden

===== 1 (0:1)

1 ===== 1 (1:1)

===== M (0:M)

1 ===== M (1:M)

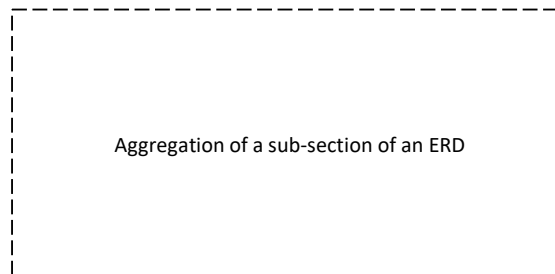
Optional

----- 1 (0:1)

1 ----- 1 (1:1)

----- M (0:M)

1 ----- M (1:M)



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